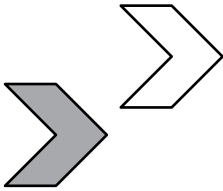


Additional Practice

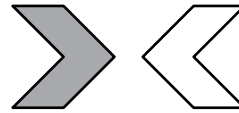
ACC7.1.01

1. Determine whether each transformation shows a translation, reflection, or rotation.

a

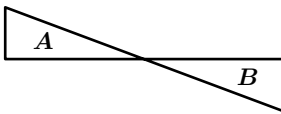


b

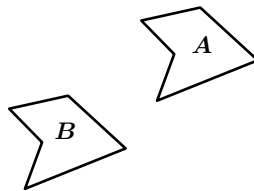


2. Determine whether the transformation from figure *A* to figure *B* shows a translation, reflection, or rotation.

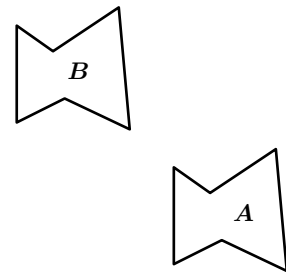
a



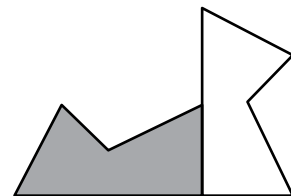
b



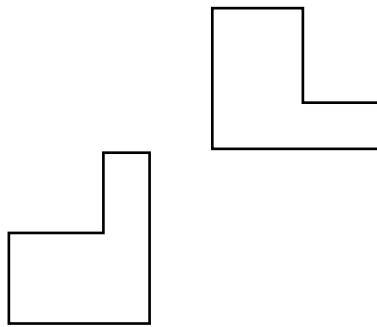
c



3. Describe the transformation.

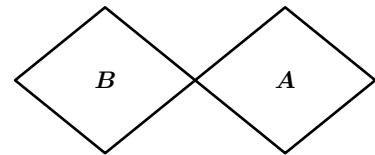


4. Describe the transformation.



5. Which transformation could move figure *A* to figure *B*?
Select *all* that apply.

- ☐ A. Reflection
- ☐ B. Rotation
- ☐ C. Translation
- ☐ D. None of these

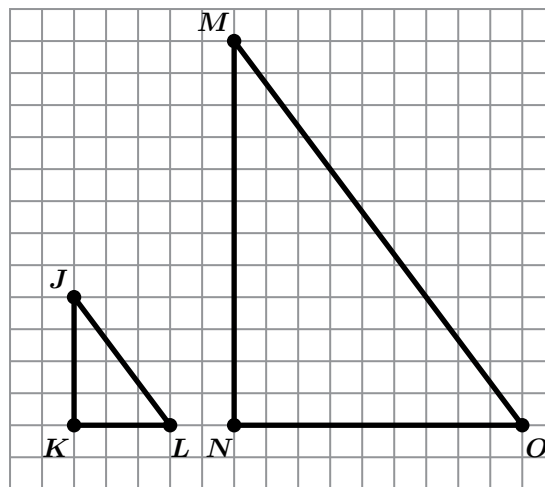


Additional Practice

ACC7.2.01

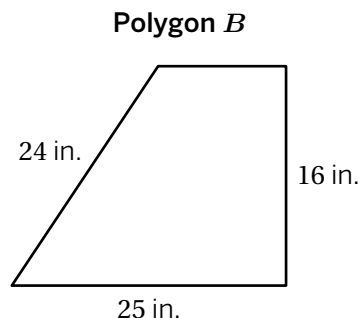
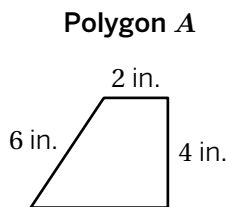
1. Triangle MNO is a scaled copy of triangle JKL . What is the scale factor that takes triangle JKL to triangle MNO ?

- A. 2
- B. 3
- C. 4
- D. 5



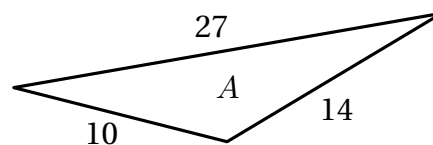
2. Polygon B is a scaled copy of polygon A .

- a What is the length of the shortest side of polygon B ?
- b What is the length of the longest side of polygon A ?



3. Figure A is scaled by a factor of $\frac{1}{2}$. Select *all* three side lengths of the scaled copy.

- | | |
|----------------------------------|--------------------------------|
| ☐ A. 7 | ☐ B. 28 |
| ☐ C. 13.5 | ☐ D. 20 |
| ☐ E. 5 | ☐ F. 54 |

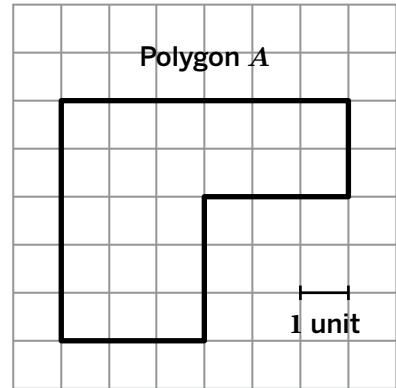


4. Polygon A has side lengths of 10, 18, 20, and 22 units. Polygon B is a scaled copy of polygon A . Which of the following sets of values could *not* be the side lengths of polygon B ?

- A. 5, 9, 10, 11 B. 25, 45, 50, 55
C. 20, 28, 30, 32 D. 2, 3.6, 4, 4.4

5. Polygon B is a scaled copy of polygon A . The shortest side of polygon B is 13.5 units.

- a What is the scale factor that takes polygon A to polygon B ?
- b What is the length of the longest side of polygon B ?



Additional Practice

1.10

Problems 1–4: James and Piper buried a treasure together on their school’s field. The field is 400 feet wide. James made a map that is 8 inches wide to record its location.

1. Write two possible scales James could have used to make his drawing.
2. Piper made her own map using a scale of 1 inch to 20 feet. Whose map is larger: James's or Piper's? Explain your thinking.
3. On Piper's map, the treasure is 2 inches from the south edge of the field. How far is the treasure from the south edge on Piper's map?
4. On Piper's map, the area of the field is 16 square inches. Piper says that the actual area of the field is 320 square feet. Is Piper correct? Explain your thinking.

5. Select *all* the scales that are equivalent to 4 centimeters to 20 meters.

- ☐ **A.** 4 inches to 20 inches
- ☐ **B.** 1 centimeter to 5 meters
- ☐ **C.** 10 meters to 2 centimeters
- ☐ **D.** 5 millimeters to 3 meters
- ☐ **E.** 1 inch to 4 feet

Explain your thinking for the scale(s) you selected.

6. On a blueprint, the living room is 4.2 inches wide. The blueprint has a scale of 1 inch to 5 feet. How wide would the living room be on a blueprint that has a scale of 1 inch to 20 feet?

- A.** 1.2 inches
- B.** 2.1 inches
- C.** 3.2 inches
- D.** 31 inches

Additional Practice

ACC7.3.02

1. Han purchased 4 packs of tomato plants. He also purchased 5 pepper plants. He bought 17 plants in all. Which equation represents the scenario?

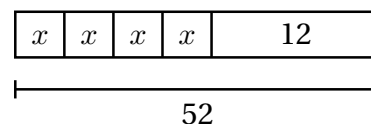
A. $17 = 4x + 5$

B. $17 = 4(x + 5)$

C. $17 - 5 = 4x$

D. $4x + 17 = 5$

2. Clare had 4 packages of stickers. After her sister gave her 12 more stickers, she had a total of 52 stickers. In the tape diagram, what does x represent?



Problems 3–5: Here are two different situations.

3. Match each equation and situation with its tape diagram.

Equation/Situation

a. $8x + 4 = 28$

b. $4x + 8 = 28$

- c. Lin's mother bought 4 kid's meals. She also bought a meal for herself that cost \$8. She spent a total of \$28 on lunch.

- d. A preschool teacher has 28 toy cars. They give equal amounts of toy cars to 8 students. They have 4 toy cars left over.

Diagram A

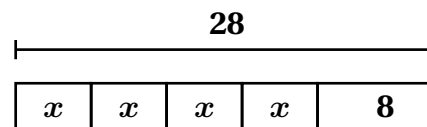
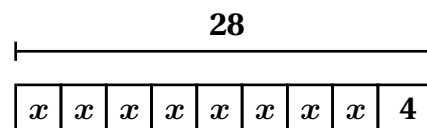


Diagram B



4. Using your responses from Problem 3, determine the value and meaning of x in Diagram A. Explain what x represents in context.
5. Using your responses from Problem 3, determine the value and meaning of x in Diagram B. Explain what x represents in context.

6. Diego and two friends each spent x dollars on a round of mini-golf. They spent \$16 on a veggie pizza. The total cost of the mini-golf and the pizza was \$34.

- a Draw a tape diagram and write an equation that represents the scenario.

- b Solve the equation. Show your thinking. What is the cost of one round of mini-golf?

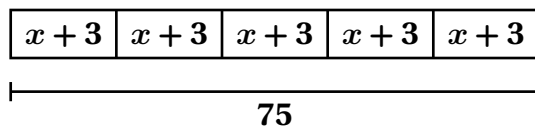
Problems 7–10: Priya wanted to run 54 miles before her next race. She has already run 18 miles and plans to run 6 more days. Priya wants to know how many miles to run each day if she wants to split up her remaining miles evenly between the 6 days.

7. Identify what the variable, x , represents in this context.
-
-
-
-
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-
-
-
-
-
8. Draw a tape diagram to represent this situation.
-
-
-
-
-
-
-
-
-
-
9. Using your tape diagram, write an equation to represent this situation.
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-
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-
-
-
10. What is the value of x ? Show or explain your thinking.

Additional Practice

ACC7.3.03

1. A kindergarten teacher placed an equal amount of stickers at 5 tables. Then he placed 3 more stickers at each table. He placed 75 stickers in all.



This tape diagram represents the situation.
How many stickers were originally placed on each table?

2. Kiran gives away a total of 28 bouncy balls to his 4 cousins. He first gives an equal amount to each cousin, and then gives each cousin 2 more. Which equation represents this situation?
- A. $6(x + 2) = 28$
- B. $2(x + 4) = 28$
- C. $4(x + 2) = 28$
- D. $6(x + 4) = 28$

Problems 3–5: A family buys 6 hats. Each family member also spends \$4 on a pair of gloves. They spend \$72 total.

3. Which equation represents this situation? Circle one.

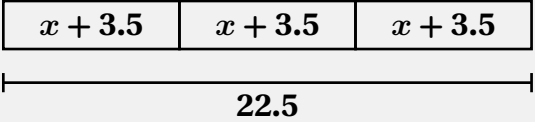
$4(x + 6) = 72$

$6(x + 4) = 72$

4. What does x represent in the equation you chose?

5. Solve the equation you chose. What does the solution tell you about this situation?

Problems 6–9: For each problem, there is a situation, diagram, or equation. Complete the table and solve each equation.

	Situations	Diagram	Equation	Solution
6.	Mai packaged 36 cans of vegetable soup in 6 boxes for a canned food drive. She first placed an equal amount in each box and then added 2 cans to each box.			
7.	A group of 4 friends each purchased a ticket to drive go-karts. Each friend also spent \$9 on a meal. They spent \$48 in total.			
8.			$4x + 5 = 17$	
9.				

Additional Practice

ACC7.3.04

1. Determine what changes were made to the first hanger that resulted in the second hanger.

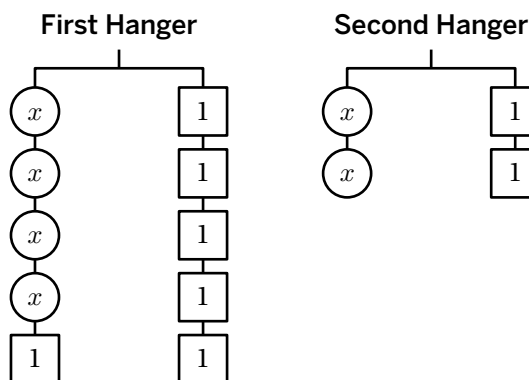
First Hanger	Second Hanger	What was done?

2. Determine the weight of each lettered shape so the hangers from Problem 1 stay balanced.

a $z =$ _____ **b** $x =$ _____ **c** $2z =$ _____ **d** $0.5x =$ _____

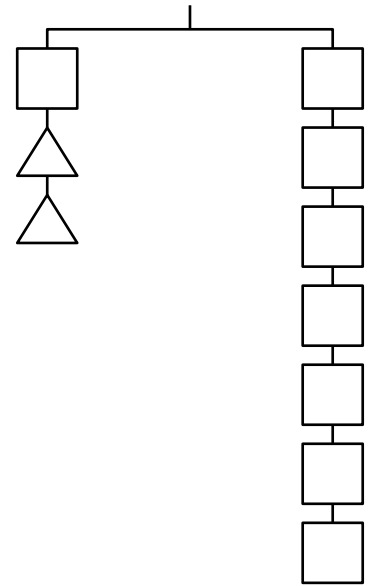
3. Here are balanced hangers. What changes were made to the first hanger that resulted in the second hanger?

- A. One unit was added to each side. Then each side was divided by 2.
- B. One unit was removed from each side. Then each side was divided by 2.
- C. One unit was added to each side. Then each side was divided by 3.
- D. Each side was partitioned into 2 equal groups, and one half of each side was removed.

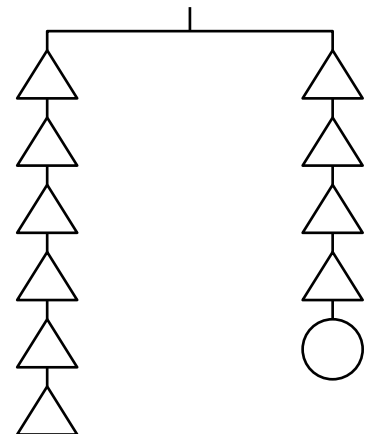


4. Determine the weight of a triangle based on these different weights for a square.

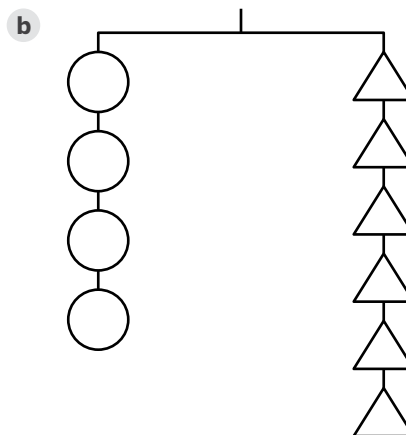
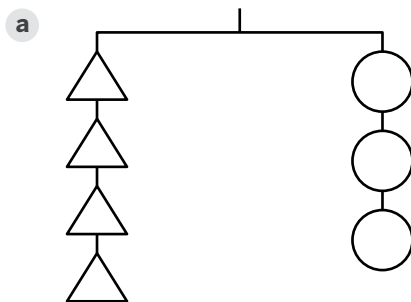
- a 2 pounds
- b 3 pounds
- c 1 pound
- d $\frac{1}{2}$ pound



5. Here is a balanced hanger. The weight of each triangle is 2.5 grams. What is the weight of the circle? Explain your thinking.



6. Determine the weight of a circle if a triangle weighs 3 pounds.

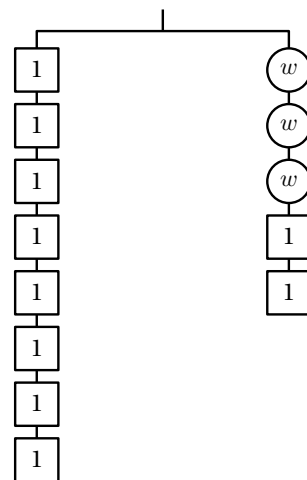


Additional Practice

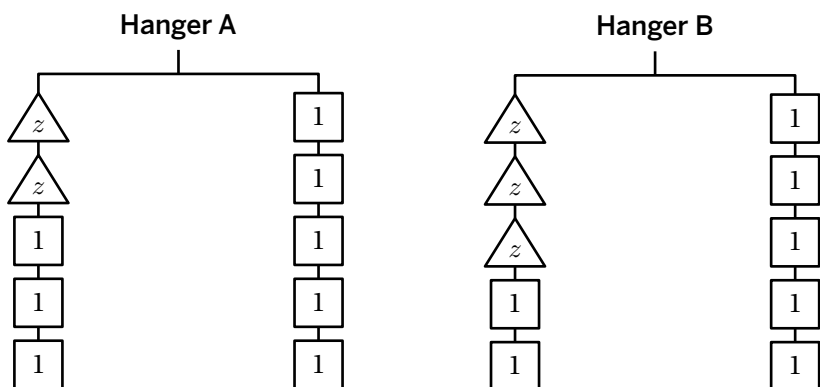
ACC7.3.05

1. Here is a balanced hanger. Which equation represents this hanger?

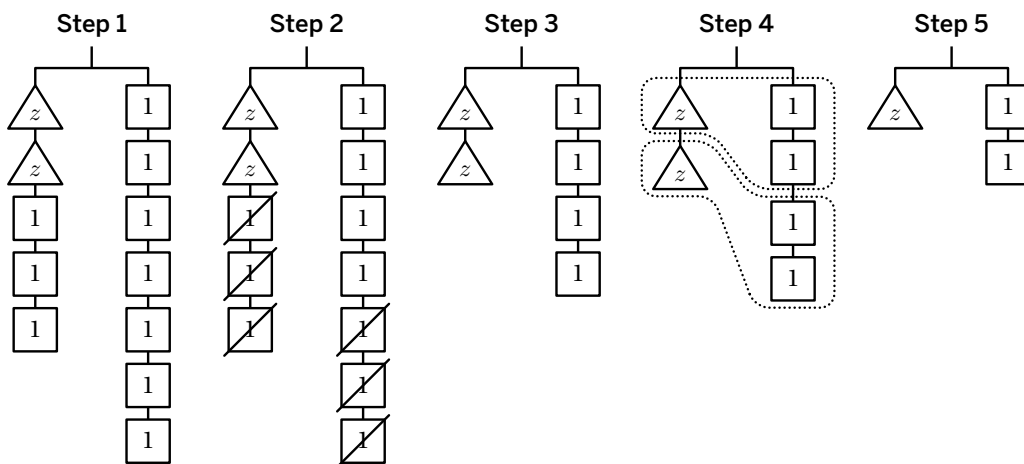
- A. $8 = 2w + 3$
- B. $10 = 3w$
- C. $8 = 3w + 2$
- D. $6 = 3w$



2. Clare described a hanger as having three unknowns and two weights balanced with five weights. Which hanger represents this description?

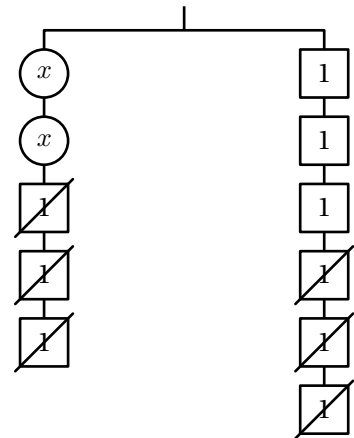


3. Andre used a hanger to find the solution to the equation $2z + 3 = 7$. Write an equation that represents each step.



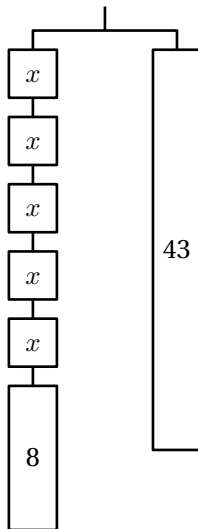
4. Bard started to solve the equation $2x + 3 = 6$ using the hanger shown. Complete Bard's work, including marking up the hanger.

Bard
 $2x + 3 = 6$
 $2x + 3 - 3 = 6 - 3$

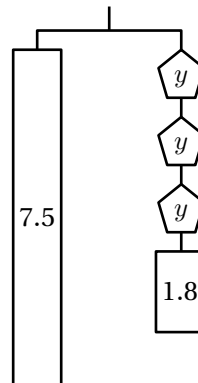


5. Solve each equation. Refer to the hanger if it helps with your thinking.

a $5x + 8 = 43$



b $7.5 = 3y + 1.8$



6. Solve each equation. Draw a hanger if it helps with your thinking.

a $y + 2 = 7$

b $2(x - 3) = 14$

Science Mom Lesson 36

Name: Date: Period:

Additional Practice

ACC7.3.06

Problems 1–3: Solve each equation by filling in the blanks.

1. $12x - 6 = 54$

$12x = \dots\dots\dots$

$x = \dots\dots\dots$

2. $5(x + 3) = -25$

$x + 3 = \dots\dots\dots$

$x = \dots\dots\dots$

3. $-75x + 150 = 0$

$-75x = \dots\dots\dots$

$x = \dots\dots\dots$

Problems 4–7: Solve each equation. Show your thinking.

4. $-10(x - 3) = 50$

5. $18 = \frac{2}{3}(w + 4)$

6. $15 = -0.5(y + 6)$

7. $-\frac{1}{3}(z - 12) = -2$

Problems 8–9: Matthew and Alice each solved the equation $-\frac{1}{3}(x + 6) = 12$ using different methods. The first steps of their methods are shown below.

8. Continue to solve the equation following each student's first steps. Show your thinking.

Matthew	Alice
$-\frac{1}{3}(x + 6) = 12$ $-3 \cdot \left(-\frac{1}{3}\right)(x + 6) = 12 \cdot (-3)$	$-\frac{1}{3}(x + 6) = 12$ $-\frac{1}{3}x + \left(-\frac{1}{3}\right)(6) = 12$

9. What is the same about each method? What is different?

10. Which method do you prefer? Explain your thinking.

Science Mom Lesson 37

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.07

Problems 1–3: Write each expression in expanded form.

1. $-3(-2)$

2. $-9(1 - y)$

3. $-14(-2y)$

Problems 4–6: Complete the missing information in the puzzle and complete the table.

4. $w - 8$

2

--	--

Factored	Expanded
$2(w - 8)$	

5.

6

$12a$	-24
-------	-------

Factored	Expanded
	$12a - 24$

6.

--	--

Factored	Expanded
	$8y - 20$

Problems 7–10: Solve each equation. Show your thinking.

7. $-3(x - 5) = 12$

8. $-4(x + 2) = 32$

9. $\frac{2}{3}(x + 3) = -18$

10. $-\frac{1}{4}(x - 7) = -10$

Additional Practice

ACC7.3.08

1. Gina says that $4x - 8$ and $3x + 4$ are equivalent because they equal 40 when x is 12. Do you agree with Gina? Explain your thinking.

2. Write at least three different expressions that are equivalent to $12w + 18$.

3. Write at least three different expressions that are equivalent to $-\frac{1}{4}(16m - 40)$.

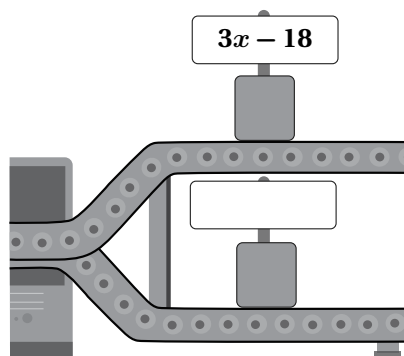
Problems 4–7: Write an equivalent expression in expanded form.

4. $-3(-a + 4) = \dots\dots\dots$
5. $-12\left(z - \frac{1}{2}\right) = \dots\dots\dots$

6. $\frac{1}{4}(20y - 10) = \dots\dots\dots$
7. $3(2x + 6y - 1) = \dots\dots\dots$

Problems 8–9: A number machine is given.

- 8.** Write an expression that will always be equal to $3x - 18$ and an expression that will never be equal to $3x - 18$. Use x as the variable in each expression.



- 9.** Explain how you know that the expressions will never be equal.

Additional Practice

ACC7.3.09

1. Select *all* of the expressions that are equivalent to $12x + 7 - 2x$.

- ☐ A. $17x$
- ☐ B. $10x + 7$
- ☐ C. $14x + 7$
- ☐ D. $12x + 7 + (-2x)$
- ☐ E. $12x - 2x + 7$

2. Fill in the blanks to make each equation true.

- a $4x + \dots = 7x$
- b $4x + \dots = x$
- c $4x + \dots = -6x$
- d $4x + \dots = 7x - 2$
- e $4x - \dots = x$
- f $4x - \dots = -6x$
- g $4x + \dots = 8$
- h $4x - (\dots) = 2x - 6$

Problems 3–6: Collect all the squares by choosing two or more expressions to combine. Then write an equivalent expression using the fewest number of terms.

All squares must be used.

$-3x$	$6 - x$	$3(x - 2)$
$2(x - 1)$	$5 + x$	$10x$
$9x + 4$	$4x$	$\frac{1}{2}x + 6$

	Original Expression	Equivalent Expression
3.		
4.		
5.		
6.		

Additional Practice

ACC7.3.10

1. Which is the solution to the equation $3x + 7 = -11$?

- A. 9
- B. 6
- C. -6
- D. -9

2. Which is the solution to the equation $\frac{2}{3}(x + 8) = -4$?

- A. 14
- B. 2
- C. -2
- D. -14

Problems 3–6: Solve each equation. Show your thinking.

3. $41 = 11 + 4z$

4. $3w + 3.1 = 9.7$

5. $-\frac{5}{4}(x - 6) = \frac{10}{4}$

6. $8000 = 4000(x - 0.06)$

- 7.** Two students are solving the equation $4(x + 6) = 22$. One student starts by using the distributive property. Another student starts by dividing both sides by 4. Show what each student's solution might look like.

- 8.** Solve the equation $-\frac{3}{4}(x - 8) = -6$ using two different methods. Show your thinking.

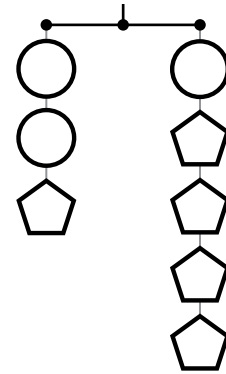
- 9.** Diego solved the equation $-3(x + 6) = 21$. His response is shown. Diego made a mistake. Identify his mistake, and then correct his response. Show or explain your thinking.

$$\begin{aligned} -3(x + 6) &= 21 \\ -3x + 18 &= 21 \\ -3x &= 3 \\ x &= -1 \end{aligned}$$

Additional Practice

ACC7.3.12

Problems 1–2: In this balanced hanger diagram, the weight of the circle is x and the weight of the pentagon is y .



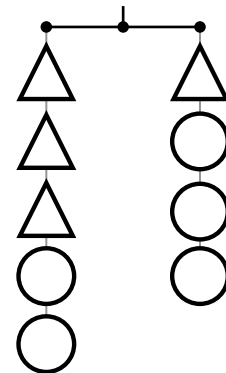
1. Label each figure with x or y . Then, write an equation using x and y to represent the hanger diagram.

2. If $x = 9$, what is the value of y ? Show or explain your thinking.

3. In this balanced hanger diagram, the weight of the triangle is x and the weight of the circle is y .

- a Label each figure with x or y .
- b Select *all* the equations that could represent a balanced hanger.

- ☐ A. $3y + 2x = y + 3x$
- ☐ B. $3x + 2y = x + 3y$
- ☐ C. $2x + 2y = 3y$
- ☐ D. $5xy = 4xy$
- ☐ E. $2x = y$



4. Match each set of equations with a possible step that turns the first equation into the second equation.

Equation	Possible steps
a. $9x + 3 = 3x + 18$ $3x + 1 = x + 6$	 Multiply each side by -3 .
b. $-3(5x - 7) = -21$ $5x - 7 = 7$	 Divide each side by -3 .
c. $-\frac{4}{3}x = 5$ $4x = -15$	 Divide each side by 3 .
d. $7x + 5 = 3x - 6$ $4x + 5 = -6$	 Subtract 3 from each side.
e. $10 - 5x = 8 + 4x$ $7 - 5x = 5 + 4x$	 Subtract $3x$ from each side.

Problems 5–6: Jada and Tyler were each trying to solve $7x - 4 = 5x - 5$. Describe the first step they each made to the equation.

5. The result of Jada's first step was $7x = 5x - 1$.
6. The result of Tyler's first step was $7x + 1 = 5x$.

Additional Practice

ACC7.3.13

1. Andre and Clare are each solving the equation $-4(x - 5) = 10x - 8$. Andre's solution is $x = -2$, and Clare's solution is $x = -\frac{7}{9}$. Their responses are shown.

Andre's solution

$$\begin{aligned} -4(x - 5) &= 10x - 8 \\ 4x - 20 &= 10x - 8 \\ -20 &= 6x - 8 \\ -12 &= 6x \\ -2 &= x \end{aligned}$$

Clare's solution

$$\begin{aligned} -4(x - 5) &= 10x - 8 \\ x - 5 &= 10x + 2 \\ -5 &= 9x + 2 \\ -7 &= 9x \\ -\frac{7}{9} &= x \end{aligned}$$

Whose solution is correct?

Andre

Clare

Neither

Show or explain your thinking.

2. Solve the equation $-5(2x - 2) = 20x$ *two* different ways. Show your thinking.

3. Mai solved the equation $\frac{1}{2}(6x - 8) = 5x + 12$. Mai began with the original equation and her steps are shown. Complete the table by describing the steps Mai used to solve the equation.

Steps	Description
$\frac{1}{2}(6x - 8) = 5x + 12$	Original equation
$3x - 4 = 5x + 12$	
$-4 = 2x + 12$	
$-16 = 2x$	
$-8 = x$	

Problems 4–5: Determine whether $x = 2$ is a solution for each equation. Show your thinking.

4. $4(x - 5) = -12$

5. $-3(x + 2) = 2x + 16$

6. Solve the equation. Show your thinking. Check your solution.

$$4(x + 3) = 6x + 2(x - 4)$$

Additional Practice

ACC7.3.15

1. For each equation, determine whether it has *one solution*, *no solution*, or *infinitely many solutions* by placing a check mark in the appropriate box.

Equation	One Solution	No Solution	Infinitely Many Solutions
$x + 2 = x + 4$			
$2x + 4 = 2x + 1 + 3$			
$3(x - 1) = 2x + 1$			
$-(4x - 5) = -x + 5 - 3x$			

2. If an equation is never true for any value of x , which is true about the equation?

- A. It has no solution.
- B. It has one solution.
- C. It has infinitely many solutions.
- D. Zero is its only solution.

Problems 3–4: For each equation, decide whether it has *one solution*, *no solution*, or *infinitely many solutions*. Show or explain your thinking.

3. $3x - 7x + 1 = -4x + 5$

4. $4(x + 3) = -2(2x + 6)$

5. Elena said $4x - 6 = 4(x - 6)$ has infinitely many solutions. Is Elena correct? Explain your thinking.

6. Noah and Lin solved the equation $-(3x + 2) - 1 = -2x - (x + 3)$. Their responses are shown below.

Noah's Work

$$\begin{aligned}-(3x + 2) - 1 &= -2x - (x + 3) \\-3x - 2 - 1 &= -2x - x - 3 \\-3x - 3 &= -3x - 3 \\-3 &= -3\end{aligned}$$

Lin's Work

$$\begin{aligned}-(3x + 2) - 1 &= -2x - (x + 3) \\-3x - 2 - 1 &= -2x - x - 3 \\-3x - 3 &= -3x - 3 \\-3x &= -3x \\x &= x\end{aligned}$$

Noah says that one of them made a mistake because you cannot get different results for the same equation. Is Noah correct? Explain your thinking.

7. Write the other side of this equation so that it is true for all values of x .

$$-\frac{1}{2}(4x + 12) + 5x = \dots\dots\dots$$

8. Write the other side of this equation so that it is true for no values of x .

$$-\frac{1}{2}(4x + 12) + 5x = \dots\dots\dots$$

9. Write the other side of this equation so that it is true when $x = 0$.

$$-\frac{1}{2}(4x + 12) + 5x = \dots\dots\dots$$

Science Mom Lesson 45

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.16

1. Without solving them, identify whether each equation has a solution that is *positive*, *negative*, or *zero* by placing a check mark in the appropriate box.

Equation	Positive	Negative	Zero
$12x = -36.24$			
$2x = 14.6$			
$6x - 10 = -10$			
$-\frac{2}{3}x = -64$			

Problems 2–5: Solve each equation. Show your thinking.

2. $5x + 7x - 12 = 6(2x - 2)$

3. $\frac{1}{2}(6y - 12) = 2(y - \frac{3}{2})$

4. $4(6 - 2m) = 3(m + 1) + 10$

5. $6b + 8 - 10b + 4 = -16 - 4b - 2$

6. Andre studied the equation $3(3x - 6) = -9(x - 2)$. He said, "I can tell right away there is no solution because on the left side, you will have $9x$, and on the right side, you will have $-9x$. $-9x$ and $9x$ make zero."

Is Andre correct? Explain your thinking.

Problems 7–8: Jada wrote the equation $8x + 2 = 8x + 4$. She wants to change only one term so that the equation has exactly one solution.

7. What is an example of a change that would result in exactly one solution?
8. What is an example of a change that would result in an *infinite number of* solutions?

Additional Practice

ACC7.3.17

1. For what value of x do the expressions $\frac{3}{4}x + 12$ and $\frac{1}{4}x - 8$ have the same value? Show or explain your thinking.

2. Which story could the equation $4x + 4 = 5x - 6$ represent?

- A. Han and his sibling get a weekly allowance of x dollars. At one point, Han had an allowance balance of $-\$6.00$ and his sibling had an allowance balance of $\$4.00$. After 5 weeks of allowance for Han and 4 weeks of allowance for his sibling, they have the same allowance balance.
- B. The Huskies and the Cardinals are playing a trivia game. Each correct question is worth x points. At one point in the game, the Huskies have -6 points and the Cardinals have 4 points. After the Huskies answer 4 correct questions and the Cardinals answer 5 correct questions, they have the same number of points.

3. At Lin's fitness center, a membership costs $\$40$ a month and there is a one-time registration fee of $\$20$. At Kiran's fitness center, a membership costs $\$50$ a month and there is a one-time registration fee of $\$10$.

Which equation represents when the costs of the membership would be equal if m represents the number of months?

- A. $40m + 10 = 50m + 20$
- B. $40m = 50m + 10$
- C. $40m + 20 = 50m$
- D. $40m + 20 = 50m + 10$

Problems 4–7: Clare and Mai are running in the same direction on the same running trail.

4. Clare runs at a constant speed of 4 miles per hour. Write an expression that shows how many miles Clare has run after t hours.
5. Mai started running $\frac{1}{4}$ of an hour after Clare. If Clare has been running for t hours, how long has Mai been running?
6. Mai runs at a constant speed of 6 miles per hour. Write an expression that shows how many miles Mai has run after Clare has been running for t hours.
7. Use your expressions to determine when Clare and Mai will meet each other on the running trail. Show or explain your thinking.
8. Shawn wants to take photography lessons. The instructor offers two options. Option 1 costs \$60 per lesson and comes with a camera. Option 2 costs \$40 per lesson but you have to purchase your own camera. Suppose Shawn buys a \$200 camera and chooses Option 2. After how many lessons, x , is Shawn's total cost the same as it would have been if he had chosen Option 1? Show your thinking.

Additional Practice

7.07

1. At a restaurant, a minimum of 2 people in each party need to be present in order to be seated at their table. Write an inequality that represents the possible number of people p that need to be present in order to be seated.

2. Clare is younger than Bard. Bard is 13 years old. Write an inequality that compares Clare's age in years c to Bard's age.

3. Diego started cooking dinner before 5:00 p.m. and finished cooking dinner at 6:00 p.m. Let d represent the number of hours Diego spent cooking dinner. Determine whether each statement is *definitely true*, *definitely not true*, or *possibly true*.

a $d < 1$

b $d > 1$

c $d < 2$

d $d > 2$

e $d < 0.5$

f $d > 0.5$

4. Noah started a run at 6:30 a.m. and finished sometime after 9:00 a.m. Let r represent the number of hours Noah spent running. Determine whether each statement is *definitely true*, *definitely not true*, or *possibly true*.

a $r > 2$

b $r > 2.5$

c $r > 3$

d $r < 3$

e $r < 2.5$

f $r < 2$

5. At the grocery store, all apples cost less than \$2.00 per pound.

- a** What is the most expensive price one pound of apples could cost?
- b** Write an inequality to represent the possible costs, in dollars, for any number of pounds of apples.

6. At the clothing store, all t -shirts cost more than \$9.99.

- a** What is the least expensive price a t -shirt could be?
- b** Write an inequality to represent the possible costs of t -shirts, in dollars.

7. Priya looks at a container of cherries and says, “I think there are more than 55 cherries in the container.”

Jada looks at the same container of cherries and says, “I think there are less than 125 cherries in the container.”

- a** Write an inequality to show Priya's statement, using c to represent the number of cherries.
- b** Write another inequality to show Jada's statement, also using c to represent the number of cherries.
- c** Can Priya and Jada both be correct? Explain your thinking and provide a possible number of cherries that supports your argument.

8. Priya and Jada are each analyzing a container of blueberries. Study their statements. Did Priya and Jada each represent their statements with correct inequalities? Explain your thinking.

Priya

I think there are less than 150 blueberries in the container. The inequality that represents this is $b > 150$, where b represents the number of blueberries.

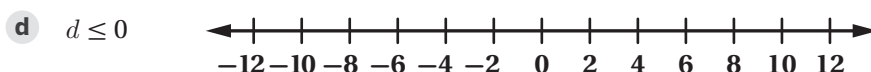
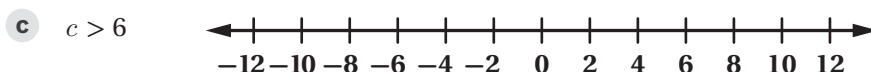
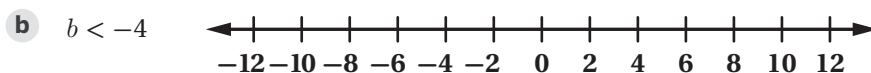
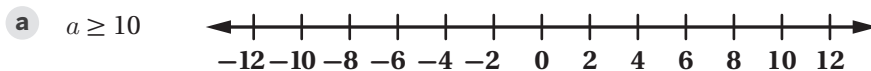
Jada

I think there are more than 65 blueberries in the container. The inequality that represents this is $b < 65$, where b represents the number of blueberries.

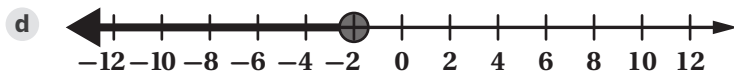
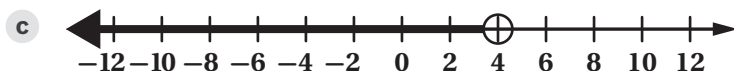
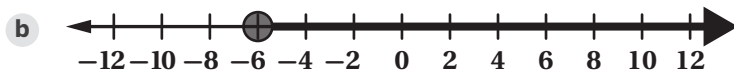
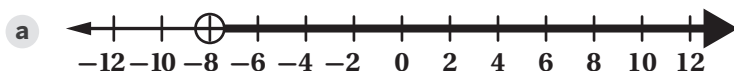
Additional Practice

7.08

1. Graph each inequality statement on the number line.



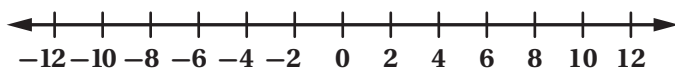
2. Use the variable x to write an inequality statement that represents each graph.



3. Consider the inequality $m < 4$.

a Which of these numbers are solutions to the inequality: 4, -3.1 , -1 , 4.6 , 1.5 , -8 , 5 ?
List *all* that apply.

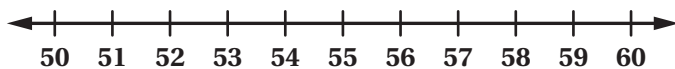
b Graph the inequality statement on the number line.



4. A sign at an amusement park reads, "You must be 54 in. or taller to ride."

a Using the variable h , write an inequality that represents the information on the sign.

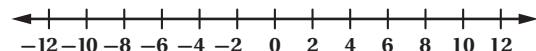
b Graph the inequality statement on the number line.



5. There are less than 12 people in line to ride the roller coaster.

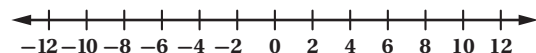
a Using the variable n , write an inequality that represents the number of people in line to ride the roller coaster.

b Graph the inequality statement on the number line.

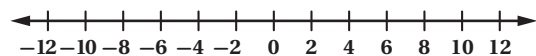


6. Represent each inequality scenario on the number line.

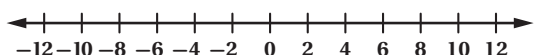
a The amusement park sells at least 10 flavors of ice cream.



b Andre spends less than \$6 at the arcade.



c It costs more than 4 tokens for the ride.



7. Andre counts more than 3 hot dog stands at the fair. He said, "To graph this scenario on a number line, I would draw a closed circle at 3 with an arrow pointing to the right." Did Andre correctly describe how to graph this scenario? Explain your thinking.

8. Priya and Jada each wrote a statement to describe a container of blueberries. Did Priya and Jada each correctly describe how to graph their inequality statements? Explain your thinking.

Priya

I think there are less than 150 blueberries in the container. To graph this inequality statement, I would draw an open circle at 150 with an arrow pointing to the right.

Jada

I think there are more than 65 blueberries in the container. To graph this inequality statement, I would draw a closed circle at 65 with an arrow pointing to the right.

Additional Practice

ACC7.3.18

1. For each scenario, circle the inequality that represents it.

- a The temperature is above 32°F.

$x > 32$ $x < 32$ $x \geq 32$ $x \leq 32$

- b Kiran has no more than \$20 to spend.

$x > 20$ $x < 20$ $x \geq 20$ $x \leq 20$

2. Consider the inequality $c \leq 9$. Select *all* the solutions to the inequality.

- ☐ A. 7 ☐ B. 8 ☐ C. 9 ☐ D. 10 ☐ E. 11

3. Write an inequality that represents each verbal description. Then write two values that will make the inequality true.

- a x is greater than 2.

- b b is less than or equal 4.5.

- c 5 is at least w .

- d d is no less than 16.

- e t is at most 8.

4. Write an inequality that represents each verbal description.

- a A student must have at least 5 hours of community service completed.

- b Practice will be no more than 50 minutes.

- c Children under the age of 3 are free.

- d During hibernation, an arctic ground squirrel's body temperature never goes below -2.9°C .

5. Consider the inequality $-2x < 14$.

- a List four values for x that would make this inequality true.
- b How are the solutions to the inequality $-2x < 14$ different from the solutions to the inequality $-2x \leq 14$?

6. Noah and Elena each wrote an inequality to represent the following situation.

A student wants to take, at minimum, \$15 to the store.

Noah's inequality: $x \geq 15$

Elena's inequality: $15 \leq x$

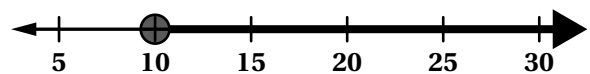
Who is correct? Circle one.

Noah Elena Both Neither

Explain your thinking.

7. In the first week of a canned food drive 480 cans were collected. In the last week of the canned food drive, at least twice that amount was collected. Write an inequality that represents the amount of cans that were collected in the last week of the canned food drive.

8. The number line shows a solution to an inequality. Write a real-world scenario that the inequality could represent.

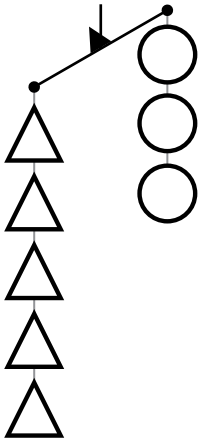


Additional Practice

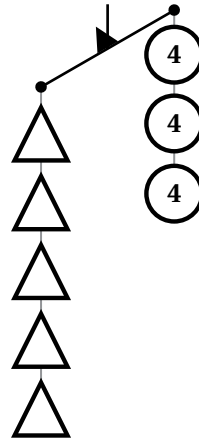
ACC7.3.19

Problems 1–2: Here are two unbalanced hangers. Write an inequality to represent the relationship between the weights on each hanger. Use t to represent the weight of the triangle in grams. Use c to represent the weight of the circle in grams.

1.



2.



3. Select *all* values of x that make the inequality $x - 7 \geq -3$ true.

☐ A. 3.9

☐ B. 0

☐ C. 4

☐ D. 4.1

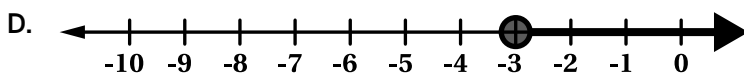
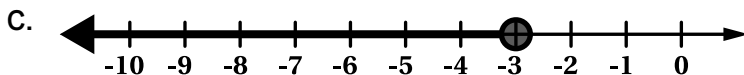
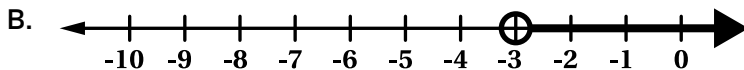
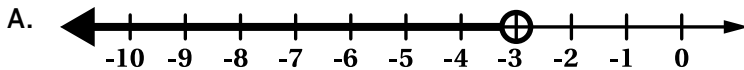
☐ E. 5

☐ F. -1

☐ G. -4

☐ H. 8

4. Which number line shows the solution to $-3x > 9$?



Additional Practice

ACC7.3.20

1. Priya has \$30 to spend at the school festival. Admission is \$4 and each ride ticket is \$2. Which inequality represents the greatest number of ride tickets she can buy?
 - A. $2n + 4 < 30$
 - B. $2n + 4 > 30$
 - C. $2n + 4 \leq 30$
 - D. $2n + 4 \geq 30$

2. It is currently 8 degrees outside, and the temperature will drop 2 degrees every hour. When the temperature falls below 0 degrees, it can be represented by the inequality $8 - 2h < 0$, where h represents the number hours the temperature has been dropping. Does $h < 4$ or $h > 4$ represent the solution of the inequality?

3. The 20 members of the photography club are trying to raise at least \$1,400 for new photography equipment. They have already raised \$540.
 - a Let m represent the amount of money each member must raise, on average, to meet their goal. Write an expression for the total amount of money going to be raised.

 - b Write an equation that represents the club raising all the money.

 - c Solve the equation. What does the solution mean in context of the scenario?

 - d Write an inequality representing the amount of money each member must raise, on average, to meet or exceed their goal.

 - e Write an inequality showing the possible average amount of money each club member needs to raise to meet or exceed their goal.

- 4.** A baker has 84 pounds of flour. After using p pounds of flour each month for 6 months, the amount of flour the baker has left is $84 - 6p$.

a Explain what the equation $84 - 6p = 76$ represents.

b What value of p makes the equation true?

c Explain what the inequality $84 - 6p \leq 76$ represents.

d What values of p make the inequality true?

- 5.** Noah spent \$40 on supplies for making 25 birdhouses. Noah wants to make a profit of more than \$300.

a Write an inequality to determine the price p Noah should charge per birdhouse if he wants to meet his goal.

b What values of p make the inequality true?

- 6.** Elena scored 95, 91, and 90 on 3 tests. She wants her average test score for 5 tests to be at least 92.

a Elena wants to know what she needs to get on her next two tests to meet her goal. Write an inequality where x represents what Elena will need to score on the next two tests.

b What values make the inequality true?

Additional Practice

ACC7.3.21

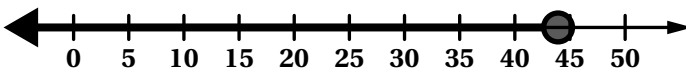
1. Select *all* the values of x that make the inequality $-x + 6 < 8$ true.

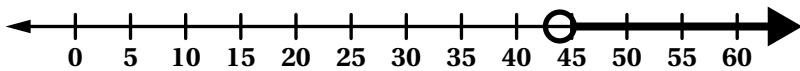
- | | |
|----------------------------------|-----------------------------------|
| ☐ A. 1.9 | ☐ B. 2 |
| ☐ C. 2.01 | ☐ D. -2.01 |
| ☐ E. -2 | ☐ F. -1.9 |
| ☐ G. 15 | ☐ H. -15 |

2. To ride on the Tall Mountain rollercoaster, guests must be at least 44 inches tall. Match each situation or graph with an inequality.

a. Kira is x inches tall and cannot ride the rollercoaster. _____ $x > 44$

b. Jake is x inches tall and can ride the rollercoaster. _____ $x \leq 44$

c.  _____ $x < 44$

d.  _____ $x \geq 44$

Problems 3–4: Here is the inequality $60 - 2x \geq -10$.

3. Select *all* the values of x that make the inequality $60 - 2x \geq -10$ true.

- | | | |
|----------------------------------|----------------------------------|---------------------------------|
| ☐ A. 0 | ☐ B. 35 | ☐ C. -35 |
| ☐ D. 35.1 | ☐ E. 34.9 | |

4. In order to solve the inequality $60 - 2x \geq -10$, Soorya solved the equation $60 - 2x = -10$ and got $x = 35$. What is the solution to the inequality? Show or explain your thinking.

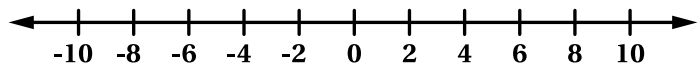
5. Jasmine is solving the inequality $12x < -48$. She solves the equation $12x = -48$ to determine $x = -4$. What is the solution to the inequality?

- A. $x < -4$
 B. $x > -4$
 C. $x \leq -4$
 D. $x \geq -4$

Problems 6–7: Complete the table to determine the solutions to each inequality. Write the solutions as an inequality and graph them on the number line.

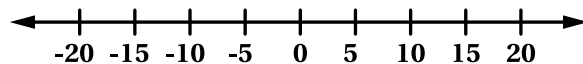
6. $-5x \geq 30$

x	-9	-8	-7	-6	-5	-4	-3
$-5x$							

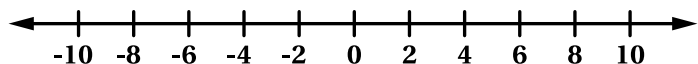


7. $\frac{4}{5}x > -12$

x	-20	-15	-10	-5	0	5	10
$\frac{4}{5}x$							



8. Solve the inequality $2x + 8 > 12$ and graph the solution on the number line. Show your thinking.



Additional Practice

ACC7.3.22

Problems 1–3: A bakery sold $\frac{2}{3}$ of the loaves of bread on display and brought out another 10. The bakery likes to keep at least 25 loaves of bread on display. The manager wrote the inequality $\frac{1}{3}x + 10 \geq 25$ to describe the situation.

1. Explain what $\frac{1}{3}x$ means in the manager's inequality.
2. Solve the inequality.
3. What do the solutions to the inequality mean in this situation?

Problems 4–6: A teacher has up to \$150 to spend on classroom supplies. He plans to purchase 12 packs of markers and plans to spend \$45.25 on posters.

4. Write an inequality to represent the situation.
5. Solve the inequality.
6. What do the solutions to your inequality mean in this situation?

Problems 7–8: Write an inequality to represent each situation.

- 7.** A library has one large meeting room that fits 20 people and many smaller rooms that hold 6 people. There are enough meeting rooms to hold at most 50 people. Write an inequality whose solution is the possible number of 6-person meeting rooms in the library.
- 8.** There are 7 recycling bins in a school cafeteria. 6 recycling bins are the same size, and the seventh holds 20 liters of recycling. Students for Environmental Action are hoping that the recycling bins can hold at least 50 liters of recycling. Write an inequality whose solution is the possible size of each of the 6 recycling bins.

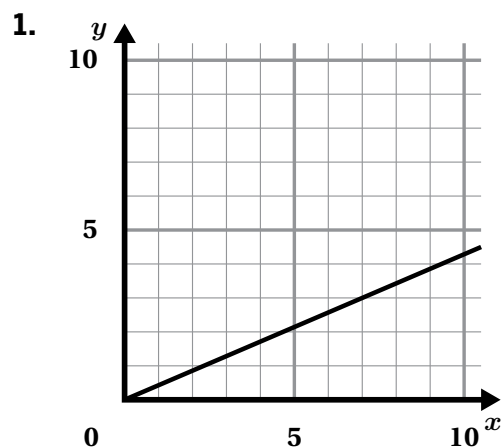
Problems 9–11: Here is the inequality $2x + 8 \leq 35$.

- 9.** Write a situation to represent the inequality. Be sure to describe what the variable represents.
- 10.** Solve the inequality.
- 11.** What do the solutions to the inequality mean in your situation?

Additional Practice

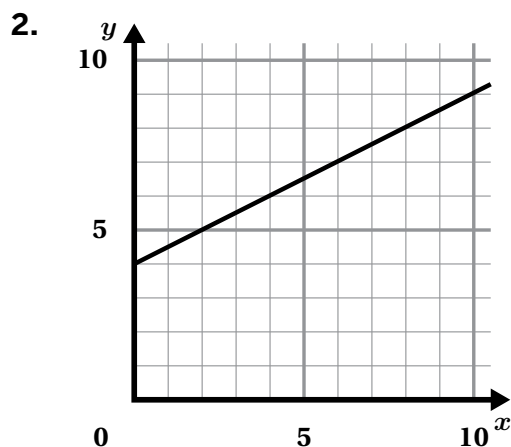
ACC7.4.01

Problems 1–4: Determine whether each graph represents a proportional or non-proportional relationship (circle one). Then, determine the slope of each line.



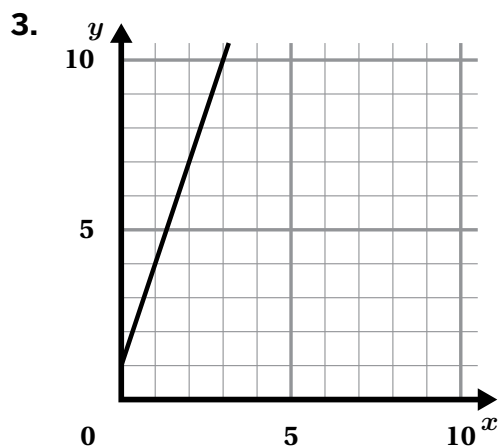
Proportional Non-Proportional

Slope =



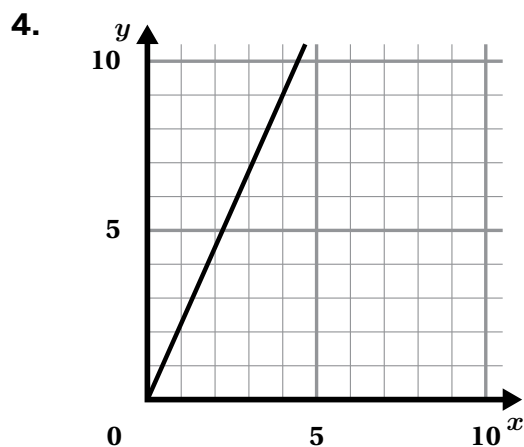
Proportional Non-Proportional

Slope =



Proportional Non-Proportional

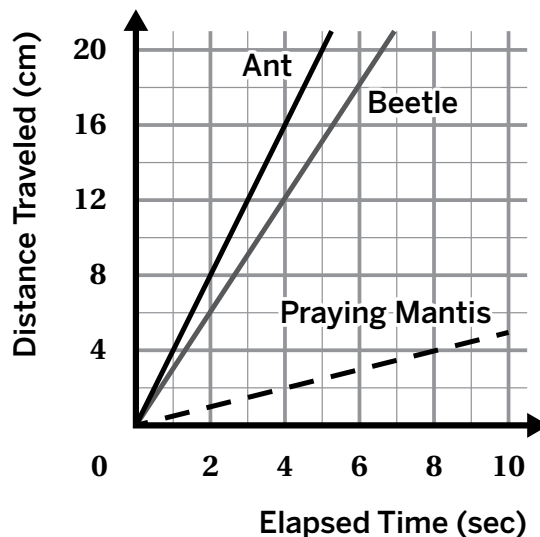
Slope =



Proportional Non-Proportional

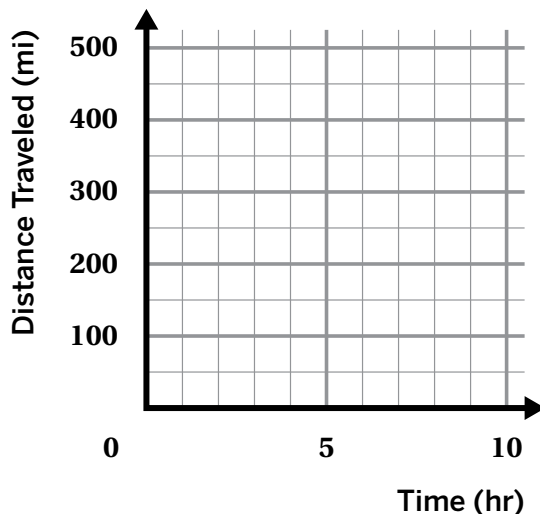
Slope =

5. An ant, a beetle, and a praying mantis compete in a 20-centimeter race. Write a story that represents the three lines on the graph.



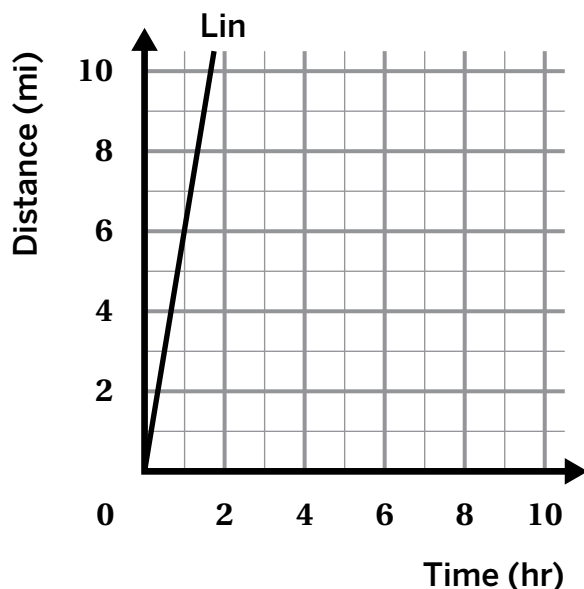
Problems 6–8: A bus begins traveling at a constant rate. After 3 hours, the bus has traveled 150 miles.

6. Graph the line showing the relationship between the car's distance traveled and time.
7. What is the slope of the line?
8. What does the slope represent in context?



Problems 9–10: Lin bikes at a constant speed. The relationship between her distance and time is shown on the graph. Jada jogs at a constant speed that is half as fast as Lin.

9. Graph the relationship between Jada's distance and time on the same coordinate plane.
10. Explain your reasoning.

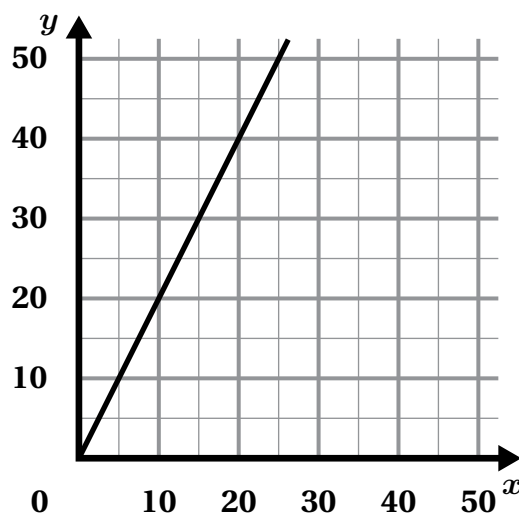


Additional Practice

ACC7.4.02

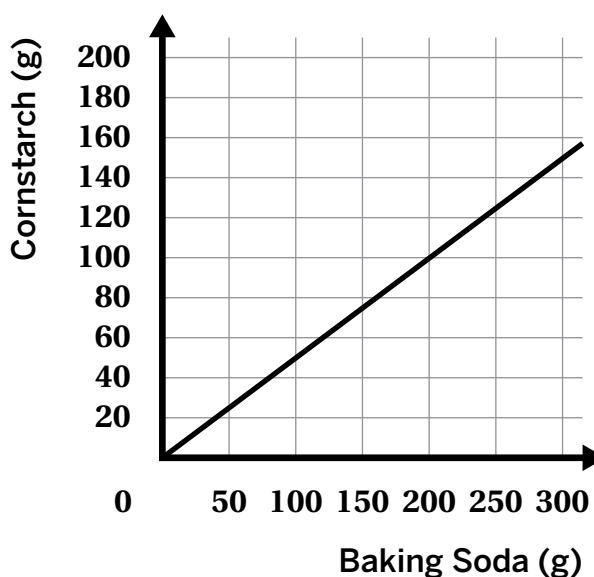
1. Write an equation for the graph of a proportional relationship that passes through the point (16, 12). Explain your thinking.

2. Write an equation of the line graphed.
Use $y = mx$ form where m represents the slope of the line.



Problems 3–4: The graph shown represents the proportional relationship between the number of grams of baking soda and the number of grams of cornstarch in a recipe to make your own bath fizzies.

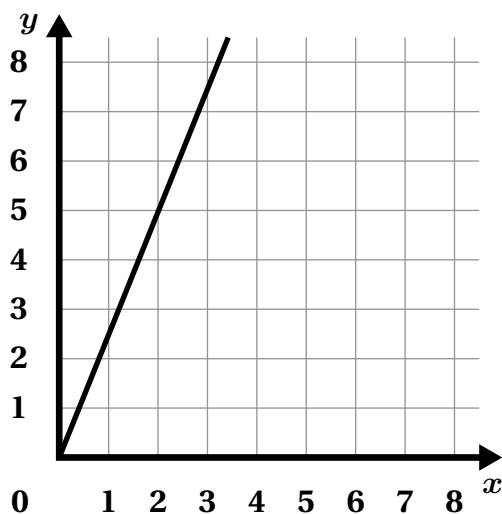
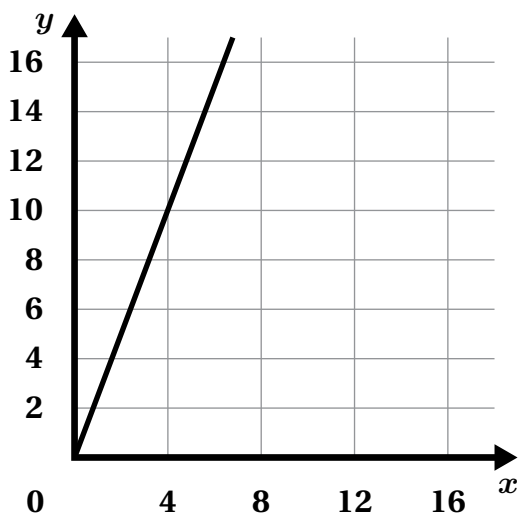
3. Write an equation that represents this relationship. Let x represent the number of grams of baking soda and y represent the number of grams of cornstarch.



4. Use your equation to complete the table.

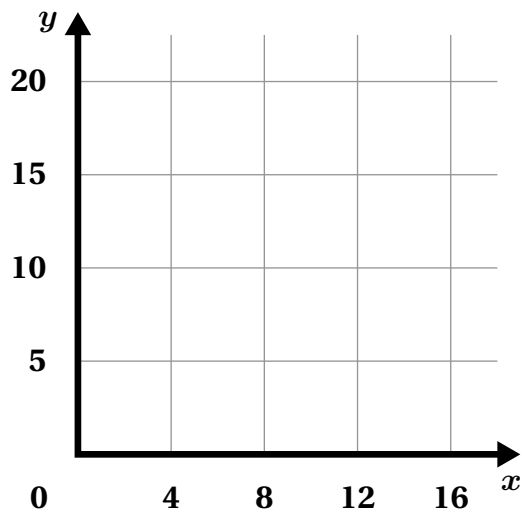
Baking Soda (g)	Cornstarch (g)
650	
	1,250
1	

Problems 5–6: A swimmer is swimming at a constant rate. The two graphs shown represent the same proportional relationship between the distance swam in feet, y , and the amount of time, x , that has passed in seconds.



5. Write an equation that represents the relationship between distance y and time x .

6. Draw a graph that shows the same proportional relationship as Problem 5. Explain your reasoning.



Additional Practice

ACC7.4.03

1. In 200 grams of beef soup, there are 80 micrograms of Vitamin A. Let x represent the amount of grams of beef soup and y represent the micrograms of Vitamin A in the beef soup. Which equations represent the relationship between grams of soup and Vitamin A? Select *all* that apply.

☐ A. $y = \frac{2}{5}x$

☐ B. $y = \frac{5}{2}x$

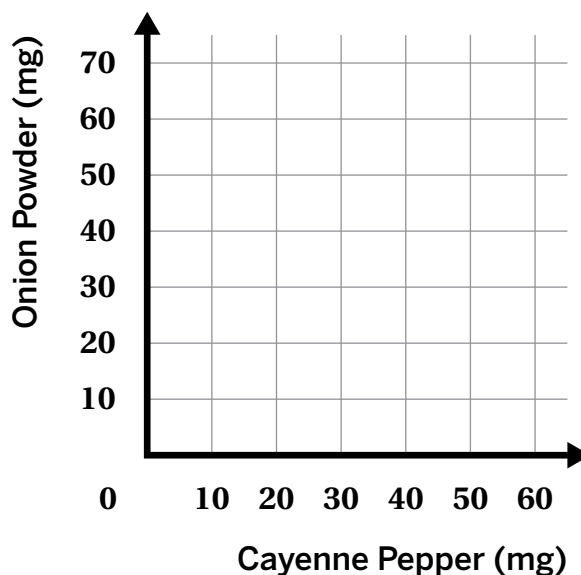
☐ C. $x = \frac{2}{5}y$

☐ D. $x = \frac{5}{2}y$

Problems 2–4: The table shows the ratios of paprika, onion powder, and cayenne pepper in a spice rub recipe.

Paprika (mg)	Onion Powder (mg)	Cayenne Pepper (mg)
60	24	16
90	36	24

2. Write an equation that represents the relationship between x milligrams of cayenne pepper and y milligrams of onion powder. Show or explain your thinking.
3. Graph the relationship on the coordinate plane.
4. How much onion powder is needed for 64 milligrams of cayenne pepper? Show or explain your thinking.



Problems 5–7: Han and Priya are at swim practice. For every 3 strokes Priya takes, Han takes 5 strokes. Suppose x represents the number of strokes Priya takes and y represents the number of strokes Han takes.

5. Complete the table.

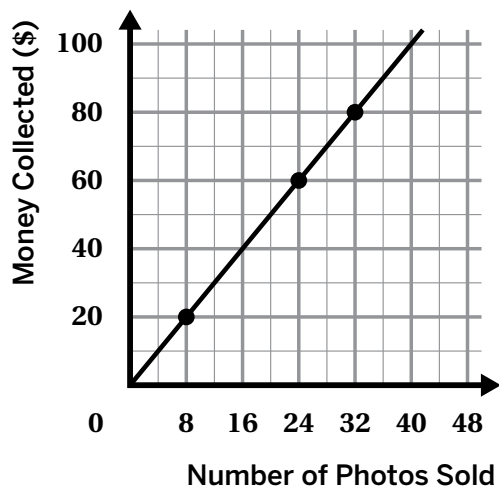
Number of Priya's Strokes (x)	Number of Han's Strokes (y)
24	
36	
75	

6. Write an equation that reflects the relationship between x and y . Show or explain your thinking.

7. How many strokes has Han taken if Priya has taken 180 strokes? Show or explain your thinking.

8. At a school festival, a photo booth is selling photos. They collect \$40 for every 16 photos they sell. Which statement correctly identifies both the slope and its meaning in the situation?

- A. The slope is $\frac{2}{5}$, so the amount of money made for every photo is \$0.40.
- B. The slope is $\frac{5}{2}$, so the amount of money made for every photo is \$2.50.
- C. The slope is $\frac{2}{5}$, so to make \$1, they have to sell 0.40 photos.
- D. The slope is $\frac{5}{2}$, so to make \$1, they have to sell 2.5 photos.



Additional Practice

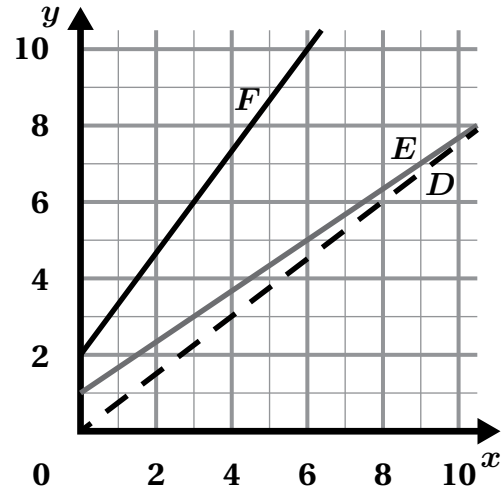
ACC7.4.04

1. Match each equation with the graph of its line.

a. $y = \frac{2}{3}x + 1$ _____

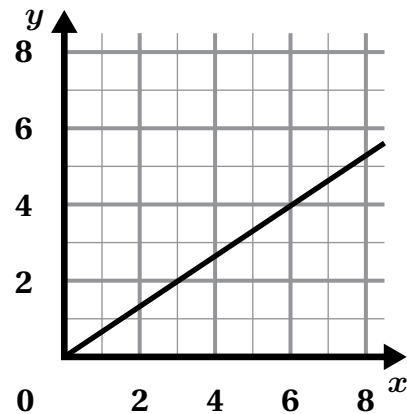
b. $y = \frac{3}{4}x$ _____

c. $y = \frac{4}{3}x + 2$ _____



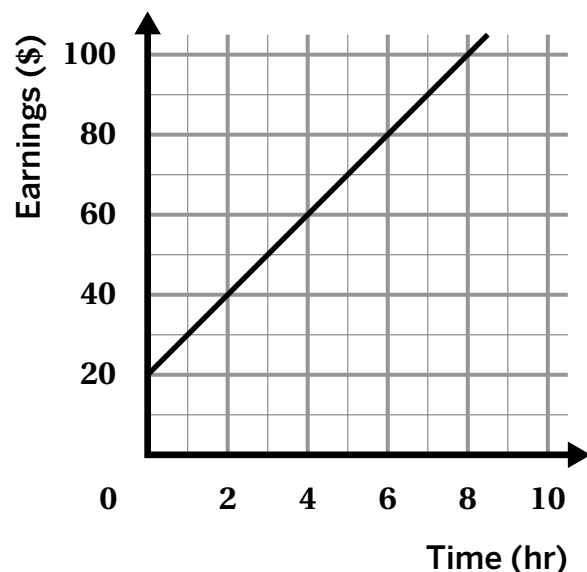
2. Select *all* statements that are true about the graph.

- ☐ A. The graph represents a linear relationship.
- ☐ B. The graph represents a proportional relationship.
- ☐ C. The constant of proportionality is $\frac{3}{2}$.
- ☐ D. The slope of the line is $\frac{2}{3}$.



3. Jada charges a one-time supply fee plus an hourly rate to do yard work. The graph shows the money she earned based on the number of hours worked for a recent all-day job.

Is the relationship a linear relationship?
Is the relationship proportional? Explain your thinking.



Problems 4–9: Determine whether these linear relationships are proportional or non-proportional (circle one).

- 4.** Noah starts biking at a constant speed of 12 miles per hour.

Proportional Non-proportional

5.

x	y
3	7
4	9
6	13

Proportional Non-proportional

- 6.** $y = 4x$

Proportional Non-proportional

- 7.** At birth, a kitten weighs 3.5 ounces and doubles its weight in the first week.

Proportional Non-proportional

- 8.** $y = \frac{1}{2}x + 3$

Proportional Non-proportional

9.

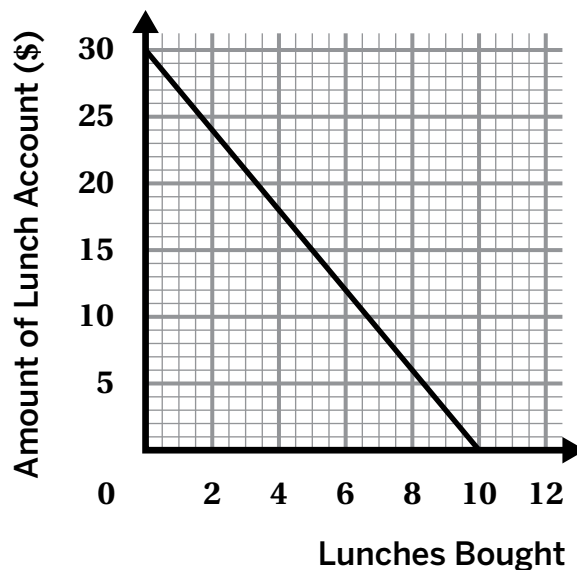
x	y
2	5
4	10
8	20

Proportional Non-proportional

Additional Practice

ACC7.4.05

Problems 1–4: Clara has an account to pay for her school lunches. Each time Clara buys lunch, \$3.00 is subtracted from the amount available in her account. This graph shows the amount of money available in her account, y , after buying x lunches.



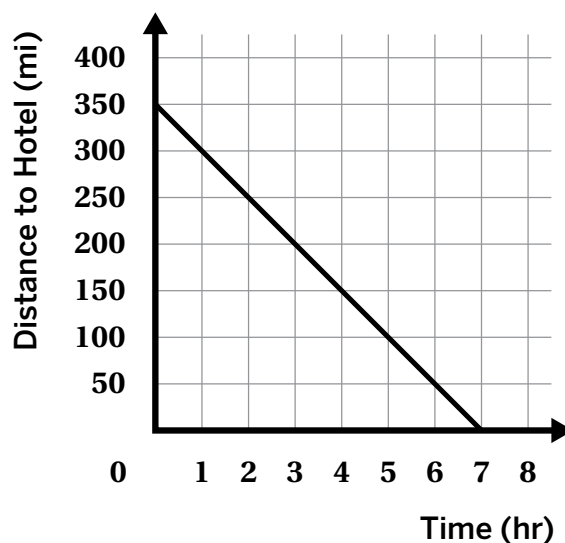
- How much money was initially in Clara's account? Explain how you know.

- Complete the table.

Number of Lunches bought	0	1	2	...	8
Amount of Money in Account		27		...	

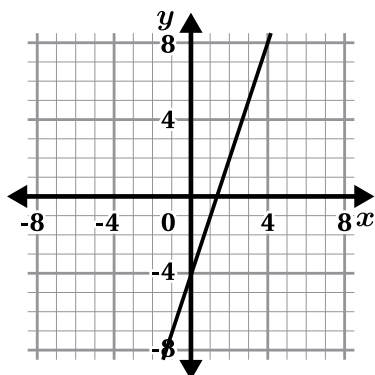
- Write an equation that represents the amount of money remaining in Clara's account, y , after buying x lunches.
- After how many lunches will Clara's account run out of money? Show or explain your thinking.

Problems 5–7: Mai’s family is driving towards a hotel from their home. Mai estimates their average speed and graphed her expected progress on the trip. The graph shows the remaining distance in miles, d , to the hotel after driving h hours.

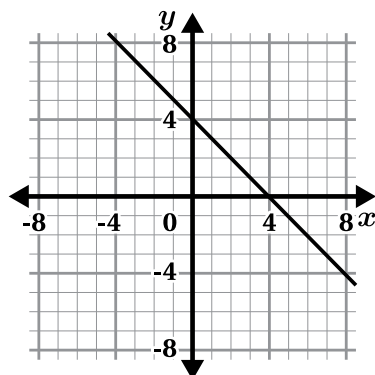


5. How far does Mai’s family live from the hotel? Explain how you know.
6. Write an equation that describes the relationship between d and h . Show or explain your thinking.
7. Approximately how long will it take Mai’s family to arrive at the hotel? Show or explain your thinking.
8. Which graph has a horizontal intercept of $(-4, 0)$?

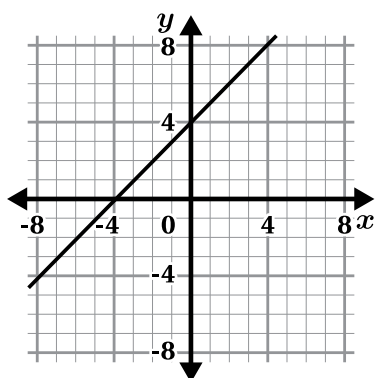
A.



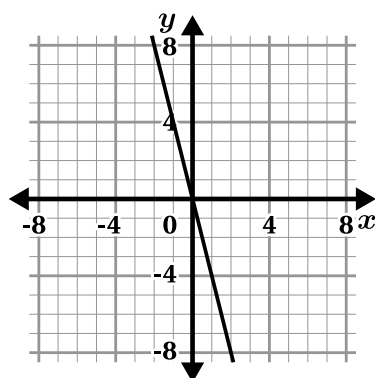
B.



C.



D.



Additional Practice

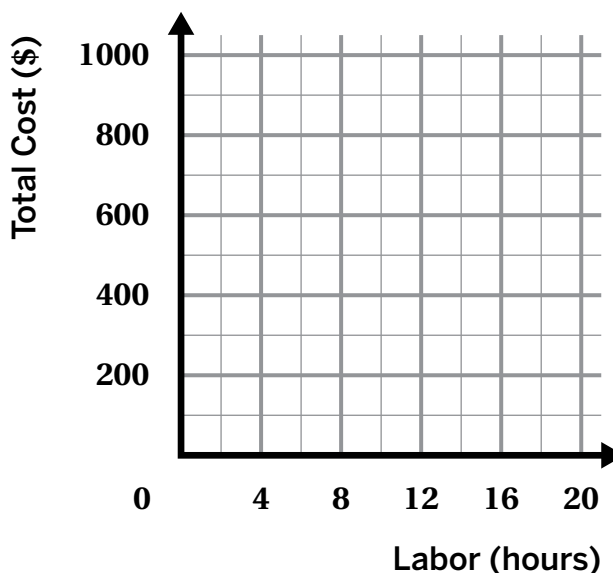
ACC7.4.07

Problems 1–4: For any service call, an electrician charges \$60, plus \$40 for each hour of labor.

- How much would the electrician charge for a service call that needs 4 hours of labor? 20 hours of labor? Write your answers in the table.

Labor (hours)	Total cost (\$)
4	
20	

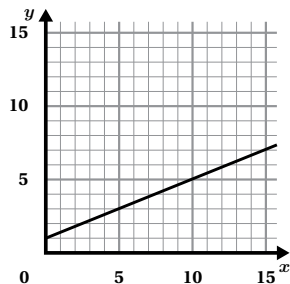
- Draw a line representing the relationship between the number of hours of labor for a service call and the total cost of the electrician's visit.



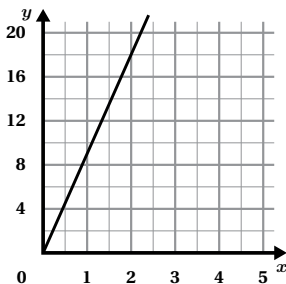
- Plot and label the points from Problem 1 on the graph.
- What is the slope of the line? What does it represent?
- A flower shop allows you to purchase additional flowers, x , to add to the total cost of your arrangement, y . The line representing the relationship between x and y has a slope of 2.75 and a y -intercept of 25.25. Explain what the slope and y -intercept represent in this situation.

Problems 6–9: For each real-world situation, choose the graph that best represents it.

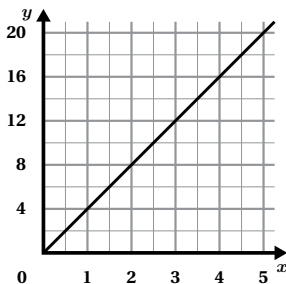
Graph A



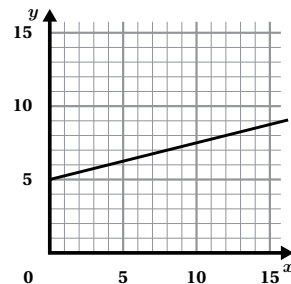
Graph B



Graph C



Graph D



6. y represents the total amount earned and x represents the number of hours worked. The slope of the line representing the relationship between x and y is 9.

Graph _____

7. y represents the total cost of a cup of ice cream and x represents the cost of each topping. The slope of the line representing the relationship between x and y is 0.4.

Graph _____

8. y represents the perimeter of a square and x represents its side length. The slope of the line representing the relationship between x and y is 4.

Graph _____

9. y represents the cost of mailing a 1-ounce package and x represents the cost added for each additional ounce. The slope of the line representing the relationship between x and y is 0.25.

Graph _____

Additional Practice

ACC7.4.08

1. Select all the equations whose graphs have the same y -intercept.

☐ A. $y = \frac{1}{2}x + 4$

☐ B. $y = \frac{1}{3}x - 2$

☐ C. $y = 4x - 2$

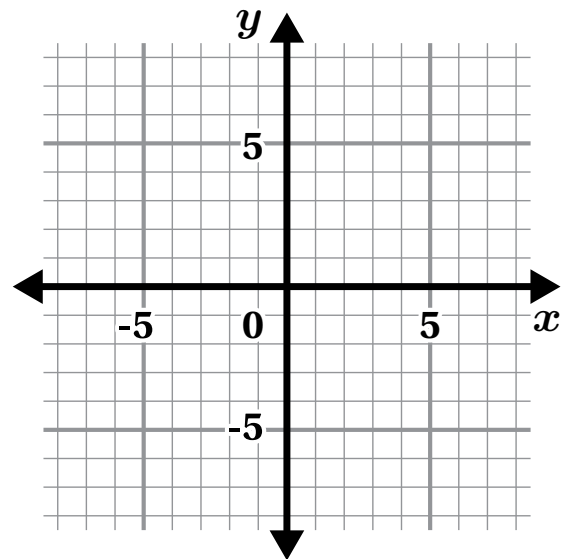
☐ D. $y = -2x + \frac{1}{2}$

☐ E. $y = \frac{1}{2}x - 2$

Problems 2–3: Here is a coordinate plane.

2. Graph the equations $y = -\frac{2}{3}x$ and $y = -\frac{2}{3}x + 4$.

3. How are the graphs alike? How are they different?



Problems 4–5: A streaming music service changes \$6 per month to existing customers. For new customers, there is an additional one-time sign-up fee of \$10.

4. Write a linear equation representing the relationship between x , the number of months of service, and y , the total amount paid in dollars by a customer.

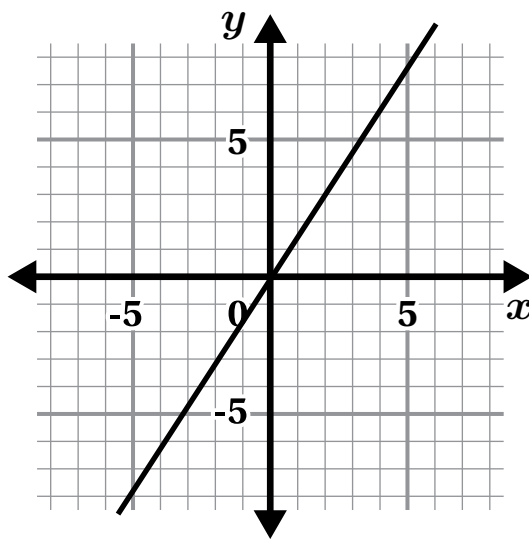
Existing customer:

New customer:

5. When the two equations are graphed on the coordinate plane, how are the graphs similar? How are they different?

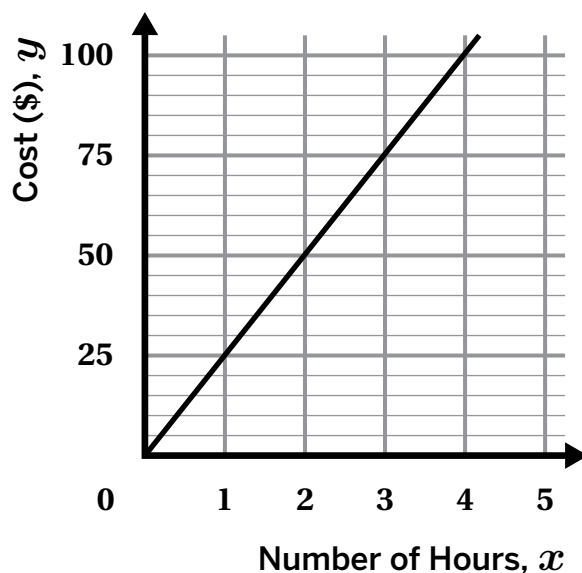
6. Here is the graph of a line. Which equation represents a line that is a translation of the line?

- A. $y = \frac{3}{2}x - 4$
- B. $y = 4x - \frac{2}{3}$
- C. $y = \frac{2}{3}x + 1$
- D. $y = -\frac{2}{3}x - 6$



Problems 7–8: A driving range charges \$25 per hour for renting a space to hit golf balls. Here is a graph that represents the cost, y , of renting a space for x hours.

- 7. Write an equation that represents the cost, y , of renting the space for x hours.
- 8. Daniel went to the driving range and also rented a bucket of golf balls for \$15. On the same coordinate plane, graph the relationship that represents the amount of money, y , that Daniel would spend after spending x hours at the driving range.

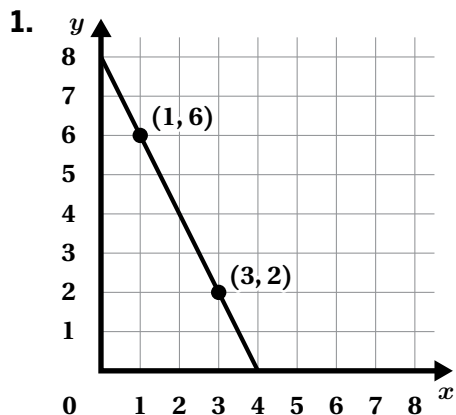


- 9. Write an equation for the relationship you graphed in Problem 8.

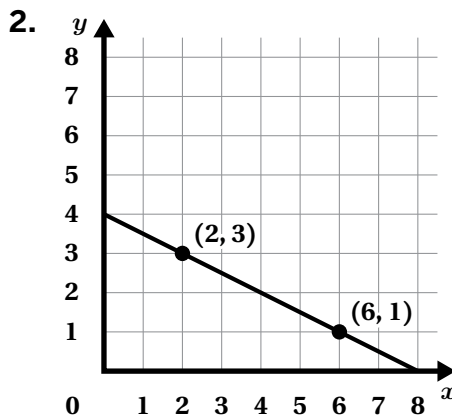
Additional Practice

ACC7.4.09

Problems 1–2: Determine the slope of each line.



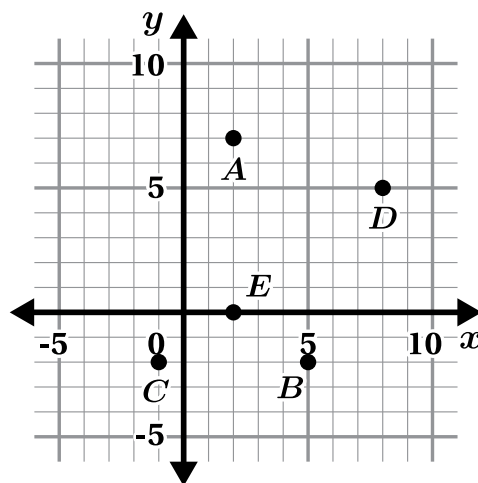
Slope = _____



Slope = _____

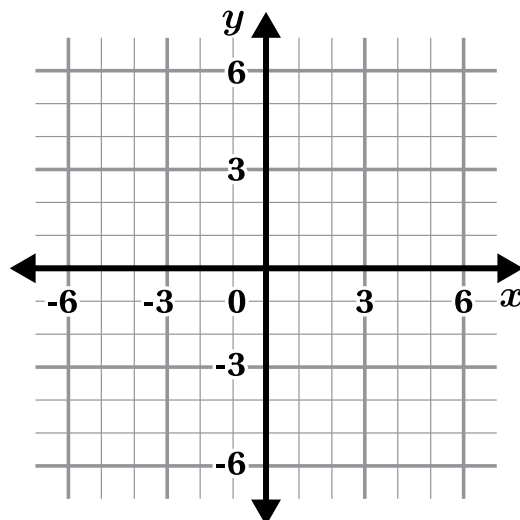
3. Draw a line with a slope of $-\frac{2}{3}$ that passes through point B .

What other point lies on that line?



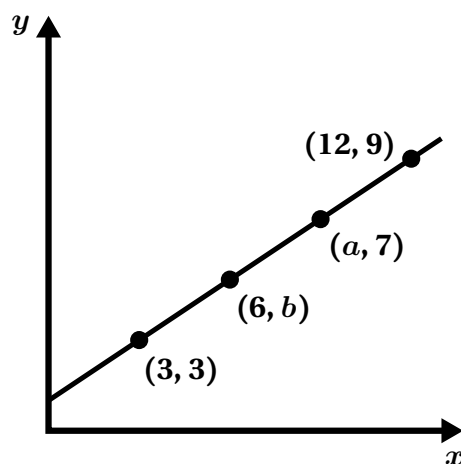
Problems 4–5: Here is a blank graph.

4. Draw a line with a slope of -3 and a positive y -intercept.
5. Explain how you know the slope of your line is -3 .



Problems 6–8: All the points in this graph are on the same line.

6. What is the slope of the line? Explain your thinking.



7. What are the values for a and b ?

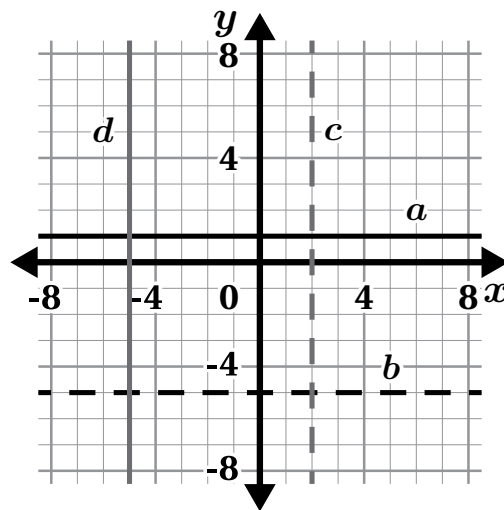
8. What is the y -value when $x = 0$?

Additional Practice

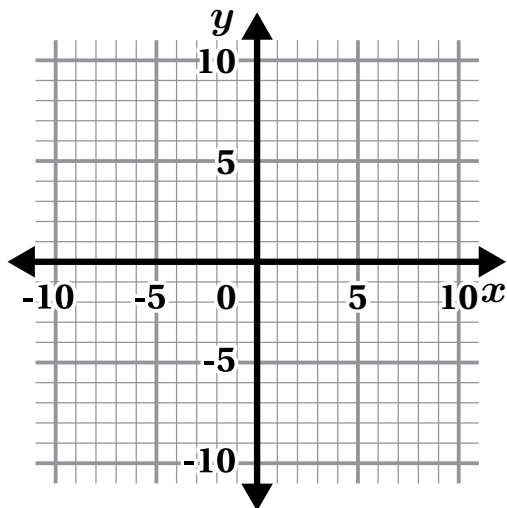
ACC7.4.10

Problems 1–4: Here are four lines on a coordinate plane. Write an equation for each line.

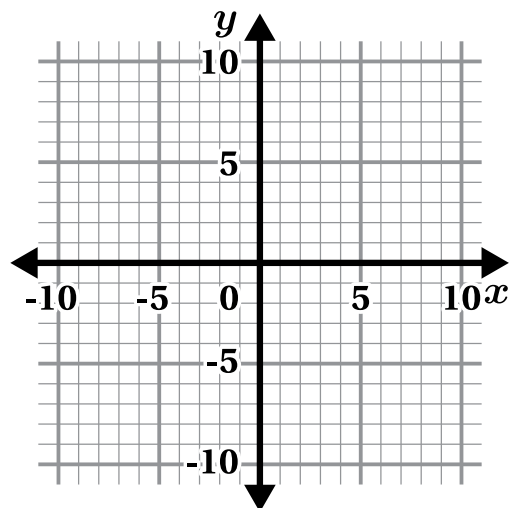
1. Equation for a :
2. Equation for b :
3. Equation for c :
4. Equation for d :



5. Draw a line with a slope of 0 and a y -intercept of 4. Then write an equation for the line.

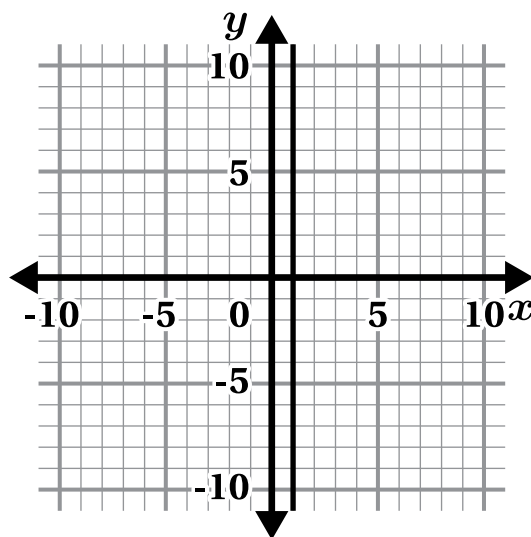


6. Draw a line with a slope of 1 and a y -intercept of 2. Then write an equation for the line.



7. This graph represents a linear relationship.
Circle the equation it matches.

- A. $y = 1$
- B. $y = -1$
- C. $x = 1$
- D. $x = -1$



8. Write an equation for the line that passes through the points (2, 7) and (5, 7).
9. Write an equation for the line that passes through the points (9, -5) and (9, 13).

Additional Practice

ACC7.4.11

1. Which equation represents the line that passes through points $(-8, 12)$ and $(4, 6)$?

A. $y = \frac{1}{2}x + 8$

B. $y = -\frac{1}{2}x + 8$

C. $y = \frac{1}{2}x - 2$

D. $y = -\frac{1}{2}x - 2$

2. Lin is writing an equation of the line that passes the point $(4, 24)$ and has a slope of 3. Her unfinished work is shown:

Lin's work:

$$y = 3x + b$$

$$m = 3$$

$$24 = 3(4) + b$$

Finish Lin's work by solving the equation for b and writing the final equation for the line in the form $y = mx + b$. Show your thinking.

3. Write the equation of the line that passes through each pair of points. Show or explain your thinking.

a $(4, 18)$ and $(8, 34)$

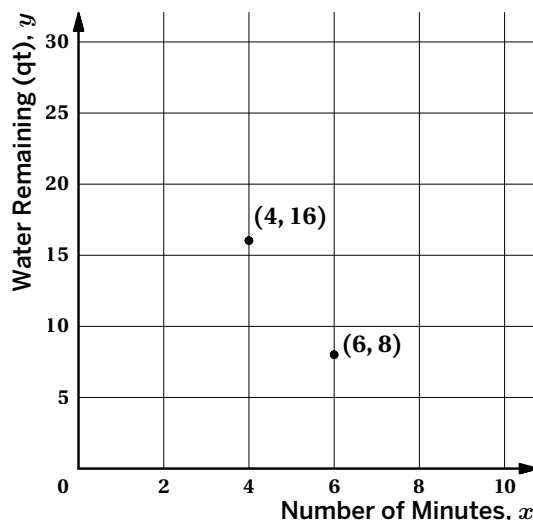
b $(-4, 14)$ and $(2, 2)$

4. Diego is finding the slope of the line that passes through the points $(10, 8)$ and $(14, 10)$. Determine any mistakes and explain how you would fix them.

Diego's work:

$$\text{Slope} = \frac{10 - 8}{10 - 14} = \frac{2}{-4} = -\frac{1}{2}$$

5. A cooler of water is draining. After 4 minutes, there are 16 quarts remaining. After 6 minutes, there are 8 quarts remaining. Write an equation for the amount of water remaining in the cooler, y , after x minutes. Show or explain your thinking.



6. The table gives the x - and y -coordinates of points on a line. Without graphing, write an equation for the line that passes through the points. Show or explain your thinking.

x	-4	4	8
y	14	2	-4

7. Consider line m with the two labeled points as shown. Kiran missed today's lesson and does not understand how he can write the equation of the line without seeing the vertical intercept. Help Kiran understand how to write the equation for the line in the form $y = mx + b$.

